



With the availability of chipsets that support both 3G and 4G technologies, the market is beginning to see 4G LTE smartphones (Image courtesy: www.digitaltrends.com)

written in small print. This included WiMAX and HSPA+ systems. There were even legal litigations in the US courts by operators against each other for false advertising. It is interesting to note that the early versions of Long Term Evolution (LTE) were being dubbed as 3.99G in published literature. ITU finally provided a more definitive definition of 4G," explains Dr Suresh Borkar, a member of faculty at the Illinois Institute of Technology, Chicago. Dr Borkar has wide consulting experience in commercial and public safety systems, telecom strategy formulation, spectrum and regulatory policies, and dynamic spectrum management.

In addition to general specifications on inter-working, handovers and quality of service, the key attributes defined by the ITU included an all-IP packet core and (downlink) bandwidth of 100 Mbps for mobile applications and 1 Gbps for nomadic and quasi-stationary applications. Taking into account the estimates and techniques like carrier aggregation and wider spectrum, there is now a general consensus in the telecommunications community that 3GPP's LTE-Advanced (Release 10) and WiMAX 2.0 (IEEE 802.16m) can be classified as 4G systems.

LTE vs WiMax— A no-brainer!

The third generation of mobile communications is characterised mainly

by HSPA+ and EV-DO. Similarly, two technologies qualify—technically—to be called as 4G. These are 3GPP's LTE-Advanced and IEEE's 802.16m (WiMAX 2.0). However, the race seems to be a no-brainer. With WiMAX 2.0 failing to take off, it looks like LTE is going to dominate the 4G world.

"WiMAX 802.16m has not been commercially successful. Hence deployment of 4G will be based on LTE-Advanced technology. These implementations are in a nascent stage worldwide. Large-scale deployment of 4G will take place only next year," says Dr Abhay Karandikar, professor and head-Department of Electrical

Engineering, IIT Bombay. IIT Bombay has made several contributions to IEEE 802.16m 4G standards, including bandwidth reservations, quality of service (QoS), relay, etc.

Experts agree that LTE-Advanced (LTE Releases 10 and 11) is the main technology on the 4G horizon. It is being standardised to use up to 100MHz channel bandwidth to achieve up to 1Gbps downlink throughput.

"In our opinion, it is okay to consider LTE (Releases 8 and 9) also to be 4G technology as it makes a radical breakthrough by adopting multi-carrier orthogonal frequency-division multiplexing (OFDM) technology. LTE has retained some level of backward compatibility to benefit from the UMTS experience and coexist with UMTS," says Harpinder S. Matharu, senior product marketing manager-Communications Division, Xilinx.

LTE technology is a completely packet-based protocol and radio access technology. Its core network, leveraging flexible bandwidth, achieves much better spectral efficiency (bps/Hz) and delivers a throughput of greater than 100 Mbps (for 20MHz FDD system using 2x2 spatial multiplexing) by using higher-order modulation and multiple-input multiple-output (downlink and uplink).

Thumbs-up 4G

1. Completely Internet protocol (IP) based architecture—a simpler, scalable, flatter and lower-cost network architecture that eases management and service deployment, and converges cellular communications with fixed-line Internet
2. Advanced and efficient air-interface based on OFDM, which provides a much higher spectral efficiency (in bps/Hz)
3. Multi-carrier OFDM modulation; multiple antennae
4. Multi-layer, multiple-input, multiple-output—downlink and uplink
5. Spectrum-agnostic design, which can support higher bandwidth, bandwidth aggregation, and time-division duplexing and frequency-division duplexing modes
6. Operates on scalable spectrum bandwidths from 1.25 to 20 MHz—the most prevalent ones being 5 and 10 MHz
7. Bandwidth of 100 Mbps for mobile applications and 1 Gbps for nomadic and quasi-stationary applications
8. Improved coverage and reliability
9. Carrier aggregation, which allows network operators to make use of several blocks of spectrum and aggregate the usage in the mobile terminal. This results in higher data rates for consumer, and enhanced spectral efficiency and increasing capacity for operators. It enables a range of wider transmission bandwidths up to 100 MHz
10. Low latency due to improved radio access protocols, and more efficient support for cloud-based services and online gaming applications
11. Self-organisation, relay and many other features to enhance spectral efficiency